

Name: _____

Class: _____

Date: _____

Unit 1 Review Packet

Creative Development



How to use this packet

Match the vocabulary, then work the mixed review questions. These are the exact words and question types you'll see on the unit test and the AP exam.

Vocab Match 1

- _____ 1. **Program**
- _____ 2. **Code segment**
- _____ 3. **Behavior**
- _____ 4. **Event**
- _____ 5. **Event-driven programming**
- _____ 6. **Iterative development**
- _____ 7. **Incremental development**
- _____ 8. **Program requirements**
- _____ 9. **Program documentation**
- _____ 10. **Comments**

- A.** how a program functions during execution; often described by how a user interacts with it
- B.** breaks the problem into smaller pieces and makes sure each piece works before adding it to the whole
- C.** a collection of program statements that is part of a program
- D.** program statements run when triggered by an event, not in a set order
- E.** an action that supplies input data to a program (click, key press, timer)
- F.** a written description of the function of a code segment, event, procedure, or program and how it was developed
- G.** requires refinement and revision based on feedback, testing, or reflection; may mean revisiting earlier phases
- H.** a collection of program statements that performs a specific task when run by a computer (also called software)
- I.** documentation written in the program that is not executed
- J.** describe how a program functions; may include descriptions of user interactions

Vocab Match 2

- _____ 1. **Logic error**
- _____ 2. **Syntax error**
- _____ 3. **Run-time error**
- _____ 4. **Overflow error**
- _____ 5. **Debugging**

- A.** the rules of the programming language are not followed
- B.** a computer tries to handle a number outside its defined range of values
- C.** an error that occurs during execution (e.g., dividing by zero, a bad list index)
- D.** the process of finding and correcting errors
- E.** a mistake in the algorithm that causes incorrect or unexpected behavior; the program runs but is wrong

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1. Testers report a confusing menu. The developer returns to the design phase to change it before continuing. This is an example of...

- A. iterative development
- B. incremental development
- C. a syntax error
- D. procedural abstraction

2. A student uses a sorting procedure found in an online tutorial in their Create task. What is the most appropriate action?

- A. Use it without mention, since it is public
- B. Use it and cite the source/original author in the program and documentation
- C. Rename its variables so it counts as original
- D. Never use outside code under any circumstances

3. In a language that stores integers using 8 bits (0 to 255), a program adds 1 to 255. What kind of error can occur?

- A. Syntax error
- B. Overflow error
- C. Logic error
- D. Roundoff error

4. A smart thermostat reads the room temperature every minute and turns on the heater if it drops below a setting. Which is an **input**?

- A. The heater turning on
- B. The electricity used by the heater
- C. The light on the thermostat's screen
- D. The room temperature reading

5. Which is the best example of program documentation?

- A. A comment explaining what a procedure does and its parameters
- B. A variable named x
- C. A faster sorting algorithm
- D. Removing unused code

6. Which is a primary reason programmers write comments?

- A. To make the program run faster
- B. To help people understand and maintain the code
- C. To fix syntax errors automatically
- D. To reduce the size of the program file

7. Which best describes the *purpose* of a program?

- A. The programming language it is written in
- B. The number of lines of code it contains
- C. The problem it solves or the creative interest it pursues
- D. The hardware it runs on

8. Which is the best example of **event-driven** behavior?

- A. A program plays a sound when the user presses the spacebar
- B. A program computes the average of a fixed list and stops
- C. A program prints the numbers 1 to 10
- D. A program sorts a list of names alphabetically

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9. Which strategy is LEAST helpful for finding a logic error?

- A. Hand tracing with a trace table
- B. Adding DISPLAY statements to show variable values
- C. Making all variable names shorter
- D. Using a debugger to step through code

10. A weather app is described: "When the user enters a city name, it displays the current temperature and a weather icon." Which TWO statements are true? (*Select two.*)

- A. The city name is an input.
- B. The weather icon is an output.
- C. The app's purpose is to display an icon.
- D. Displaying the temperature is an event.

11. This code should count values **greater than 10**:

```
count ← 0
FOR EACH val IN myList
{
  IF (val ≥ 10)
  { count ← count + 1 }
}
```

Which test list reveals the error?

- A. [10, 4, 2]
- B. [12, 15, 20]
- C. [5, 8, 3]
- D. []

12. A program runs without crashing but displays the wrong average. What type of error is most likely?

- A. Syntax error
- B. Run-time error
- C. Logic error
- D. Overflow error

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Challenge Questions

Challenge 1.2

Pick an app you use daily. In complete sentences, describe its **purpose**, one of its **functions**, one **input that is an event**, and one **output** — exactly the way you would on the Create Performance Task written response.

Challenge 1.3

Explain how investigating users *early* (surveys, interviews, observation) can reduce the amount of iterative revision needed *later*. Then describe one thing that investigation might still miss.

Challenge 1.4

A teen-club sign-up app should accept ages **13 through 18**. Explain why testing only “typical” ages like 15 is not enough, then list four test inputs you would use and the expected result for each.
